Plant Station R18 - Test and Acceptance Sheets

Controlled field	Value
Document number	PLANT-TS-001
Plant design revision	R18
Document issue	D03
Issue date	11 October 2026
Preceding document	PLANT-TS-001 R17 D02
Status	Issued for combined prototype review; physical acceptance open

Current mechanical configuration: R18 open-route tower and R18 fascia with auxiliary holes closed; retained R16 lid (R15 geometry) and R13 planter. Electrical reference remains frozen Uno UI v6 and the 27 September interface record. Arduino/relay registration is Reported; physical retention, wiring, wet and operational acceptance remain open.

R18 combines the open pump-wire route and fascia aperture closures, retaining the R17 ports, Uno/relay mounts, lid, planter and Uno UI v6. Previous issues are retained; reported board fit is not acceptance of the new parts.

Revision control

Narrow parallel left-margin bars identify surviving changes against the verified matrix. Dates are documentary dates. R15/R16 source changes are incorporated without creating intervening manual issues.

Revision / class	Change and affected content
R5-R13 / source history	Earlier assembly and mechanical changes remain traceable through the preserved D01 matrix. Missing document editions are not invented.
TS family / D02	PLANT-TS-001 assigned for this test-sheet set. Prior tests are source evidence; no earlier same-family controlled test-sheet issue is asserted.
R15 / 27 Sep 2026	Reinforced Uno bosses, thicker recessed-head lid and additional pump relief; source changes incorporated at R17, physical trials open.
R16 / 27 Sep 2026	Four relay bosses below Uno; source change incorporated at R17; actual underside and hole pitch unmeasured.
UI v6 / 27 Sep 2026	Three buttons and integrated lamps, enabled active-HIGH relay, 1-4 s request, soil display and 60 s backlight timeout. Firmware revision is independent of mechanical R17.
R17 / 28 Sep 2026	Mounted USB-C cut and pilots, adjacent diameter 10 DC extension hole; estimated dimensions authorised; coupon fit required.
D02 / 28 Sep 2026	Integrated current R17 document issue. Uses the established enhanced template, controls and cumulative revision bars. No R15/R16 manual issues created.
R18 / 11 Oct 2026	Combined open pump-wire route and closure of three standalone LED holes plus the speaker hole. Existing illuminated-button interfaces, board mounts and retained lid/planter remain. User reports Arduino and relay fit; independent screw, underside and route acceptance remain open.
D03 / 11 Oct 2026	Next controlled issue after R17 D02. Combined R18 files, partial reported fit evidence and new A18 acceptance records incorporated; A17 actions remain open. No new firmware, physical pass or accepted printer job is asserted.

D03 / document control: combined R18 technical content in the established A4 template. Pagination, new headings and editorial reflow alone receive no technical bar.

R5 is the earliest located assembly guide. R1-R4 and R9-R12 PDFs were not located in the previous inventory. R15/R16 are evidenced mechanical prototype changes, incorporated here without fictitious intermediate PDF/manual editions. Margin labels identify documentary changes, not physical test passes.

List of effective pages

All pages are effective at D03, dated 11 October 2026, for the R18 prototype. Section references provide navigation; technical change labels are controlled separately.

Page	Issue	Content
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Section / page cross-reference

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01 Test article, prerequisites and recording rules

R18 R17

Test article: combined R18 tower_open_pump_route_r18 and fascia_no_aux_apertures_r18, retained R16 lid/R13 planter, and unchanged Uno UI v6. Kevin reports newest-tower Arduino/relay fit; this is partial evidence only. No R18 physical pass is pre-populated. Link every action to actual build/part hashes, component identities, operator/date and retained evidence.

Record field	Actual value
Build ID / serial / operator / date	_____
Printed-part hashes / document issue	_____
Printer / material / slicer / orientation	_____
Component models/revisions and photographs	_____
Screw measurements / engagement	_____
Supply identifiers / ratings / protections	_____
Instruments and calibration/check status	_____
Test method / environmental conditions	_____
Deviation IDs / defect report / evidence location	_____

CAUTION
Before electrical work disconnect every supply. Before powered logic/upload checks confirm the separate pump supply is OFF. No upload, printer operation or powered test is executed by this document.

Use Pass / Fail / Not tested / Deferred with evidence. A blank actual result remains Not tested. Acceptance tolerances not supplied must be agreed and recorded before assessing that test; do not invent acceptable duration, volume, torque or leak rates.

02 Mechanical coupons and dry assembly

Action ID	Test	Expected acceptance	Existing status
A17-01	USB/DC fit coupon	USB shell and screw pattern fit; plug engages; DC thread/nut span 6 mm; no cracks	Not tested
A17-02	Uno screws and board	2.8 pilot trial takes actual screws; all four centres register without PCB bowing	Open; board fit Reported, screws/underside unaccepted
A17-03	Flush lid	Heads sit flush/below surface; pockets intact; lid seats; engagement verified	Not tested
A17-04	Pump route	Actual pump/tube clear relief; no kink; dry partition preserved	Not tested
A17-05	Relay fit	Four-hole registration; solder, screws, terminals and service removal clear	Open; board fit Reported, underside/wires unaccepted
A17-06	Fascia and full dry build	LCD/buttons/pot fit; wiring untrapped; planter lifts clear; all fasteners retain	Not tested

R18

R18

R18

R17

A17-01: print the port coupon in the intended material; measure both mounting centres and each cut; trial connectors, fasteners, actual plugs, DC nut and solder/wire access together.

R17

A17-02: trial actual screws in full-length 2.8 mm bosses; measure engagement and remaining pilot-depth margin. Inspect splitting/retention and all four Uno holes before approving a body change.

R17

A17-03/04: trial actual lid heads and pump with attached tube against matching coupons, then check seat, clearances, removal and dry boundary in the body.

R18

R17

A17-05/06: retain the reported Arduino/relay fit as an attributed observation, not a complete pass. Measure relay pitch/underside, fit all screws and inspect actual wires/tool access. Dry-build R18 fascia, boards, wiring and planter; record full removal travel. New route/fascia results also require A18-01 to A18-04.

Action ID	Actual result / evidence	Verdict / operator / date
A17-01	_____	Not tested / _____
A17-02	_____	Not tested / _____
A17-03	_____	Not tested / _____
A17-04	_____	Not tested / _____
A17-05	_____	Not tested / _____
A17-06	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action: _____. Retest configuration/evidence: _____.

03 Power interface and relay reset acceptance

Action ID	Test	Expected acceptance	Existing status
A17-07	USB/barrel electrical interface	Exact USB pins/CC/power/data, DC polarity/voltage and source behaviour identified	Open design hold
A17-08	Relay default OFF	Coil/input/contact ratings recorded; OFF through reset and both supply sequences	Not tested

R17 A17-07: identify USB-C part/pins and intended power/data adaptation; inspect configuration-channel provision. Measure actual DC extension polarity, mating dimensions and grip. Freeze the approved input diagram and source selection before energising it.

R17 Record USB input, board rails and barrel input voltages/current under the agreed load. Record the intended supply arrangement and unwanted coupling tests appropriate to it. Define numerical pass limits from matching component specifications first.

R17 A17-08: with pump disconnected, identify coil/input/contact ratings and measure contact state through board reset, open logic input and both source power-up orders. Expected contact state is OFF throughout inhibit conditions; stop on any transient closure.

R17 Verify an accessible physical pump-power disconnect, selected fuse/wire/contact ratings and motor suppression. Do not substitute a software OFF message for measured contact state.

Action ID	Actual result / evidence	Verdict / operator / date
A17-07	_____	Not tested / _____
A17-08	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action: _____. Retest configuration/evidence: _____.

04 LCD, controls and dose measurements

Action ID	Test	Expected acceptance	Existing status
A17-09	LCD and controls	Display stable, buttons/lamps verified, dose return and full 60 s timeout observed	Partial serial record only
A17-10	Dose and volume	Actual contact/pump durations and volume measured at selected endpoints	Not measured

R17 A17-09: with pump supply OFF, verify each switch and lamp; stable LCD status 0 and visible pages; STOP/release lockout; 1-4 s indicated endpoints; safe soil-index/fault test inputs; no automatic request from moisture.

R17 Observe and record the actual LCD backlight after 60 s idle, button/knob wake and 60 s after completed/cancelled requests. Confirm immediate soil Status return at request end. Retain video/timed observation and serial as separate evidence.

R17 A17-10: after power/reset protections pass, run a supervised measured pump trial using the approved supply and installed tube/lift. Measure contact ON duration and delivered volume at 1.0/4.0 s settings and intermediate points required by the intended use.

R17 Timing and volume acceptance tolerances: Not yet defined. Record instrumentation resolution, repetitions, individual readings, mean/range and failures. Do not mark a test Pass without the pre-set numerical criteria.

Action ID	Actual result / evidence	Verdict / operator / date
A17-09	_____	Not tested / _____
A17-10	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action: _____. Retest configuration/evidence: _____.

05 Leak/fill, print configuration and CAD exception

Action ID	Test	Expected acceptance	Existing status
A17-11	Leak/fill and wet route	Leak, drain-back, tilt/splash evidence; fill limit below unsealed openings recorded	Not tested
A17-12	Material/print configuration	Material, slicer layers, orientation, local fit/strength recorded	Unknown
A17-13	CAD shutdown anomaly	Clean CAD process log or isolated dependency defect with retained evidence	Open; build exits 1

R17

A17-11: define leak-test duration, fill candidate, containment and acceptance first. Record actual water level, drain-back headspace, soil/drain condition, planter movement, tilt/splash exposure and dry-tower inspection. Fill mark acceptance needs physical evidence.

R18 R17

A17-12: record actual material, printer, saved R18 job, part hashes, orientation/supports and measured holes; inspect complete layers and rear bosses. Prior R17 plate projects are historical. Candidate slice comparisons carry warnings and do not close this action or supply accepted R18 machine files.

R18 R17

A17-13: retain build outputs and exit statuses. Original candidate CAD runs report geometry PASS then exit 1. Scratch R18 regeneration reproduces all three STL hashes, but both child processes exit 3221226356 (0xC0000374) after PASS; wrapper exit 1. Combined STEP checking saves PASS then exits 1; mesh-only checks exit 0. Preserve the exception until a clean process or isolated dependency-defect evidence closes it; no physical fit follows.

Action ID	Actual result / evidence	Verdict / operator / date
A17-11	_____	Not tested / _____
A17-12	_____	Not tested / _____
A17-13	_____	Not tested / _____

Owner: Unassigned. Target date: Not set. Deviations and corrective action: _____. Retest configuration/evidence: _____.

06 R18 route, fascia and combined-build acceptance

Action ID	Test / method	Expected acceptance	Existing status
A18-01	Terminated pump-lead lay-in	Actual bundle dimensions recorded; lays into front/top channel with terminations retained and no forcing	Open / Not tested
A18-02	Wire bends, edges and retention	Deburred edges; no insulation damage/kink; qualified bend/retention basis; wire stays seated through lid/planter service	Open / Not tested
A18-03	Replacement fascia face and controls	4 positions continuous/flat without cracks; buttons/LCD/knob/fixings seat and remove without force	Open / Not tested
A18-04	Combined print and full dry build	Actual R18 job/layers/part hashes recorded; existing tube, boards, terminals, lid and planter clear; service removal untrapped	Open / Not tested

A18-01: with every supply disconnected, measure the actual bundle and termination envelopes. Remove lid/planter and lay the intermediate wire into the open front channel x106.5-114/y82-96/z52-129 and upward-open crossover x89-114/y88-96/z77-159. Keep terminations fitted. Retain measurements and photographs; optional route_trial_r18 only checks local width/lay-in.

A18-02: inspect printed edges and document the actual cable bend/strain-relief basis before assessing retention. Fit the retained lid, planter and fascia, observe pinching/lifting and repeat removal/reinstallation without disturbing boards or removing terminals. Record any protection/retention correction explicitly; do not silently add a new printed part. Stop on insulation damage, sharp contact, forcing or unrestrained wire.

A18-03: inspect the three former 5 mm LED positions and one former 4 mm speaker position through the 3 mm face plate. Record flatness/gaps/delamination. Trial the three illuminated buttons, LCD, knob, 3.4 mm fascia fixings and display bosses; rear collars/piezo seat are retained blind pockets. D10 is unused. This dry fit does not test lamp circuitry.

A18-04: review the actual combined R18 tower/fascia job for complete plate continuity, remaining supports, orientations, critical interfaces and imported units. Resolve warnings before accepted manufacture. Record printer/material/process, critical dimensions, STL hashes and full dry-assembly photographs, including water tube, relay terminals, service loops, screw access and planter lift. No numerical strength or fit tolerance is invented.

Action ID	Actual result / evidence	Verdict / operator / date
A18-01	_____	Not tested / _____
A18-02	_____	Not tested / _____
A18-03	_____	Not tested / _____
A18-04	_____	Not tested / _____

Owner: Unassigned for all A18 actions. Target dates: Not set. Test article/hashes: _____. Agreed criteria and instrument basis: _____. Deviations: _____. Retest evidence: _____.

07 Close-out and disposition

Action	Result	Evidence / owner / target
A17-01	Not tested / Open	_____
A17-02	Open; registration Reported only	_____
A17-03	Not tested / Open	_____
A17-04	Not tested / Open	_____
A17-05	Open; registration Reported only	_____
A17-06	Not tested / Open	_____
A17-07	Not tested / Open	_____
A17-08	Not tested / Open	_____
A17-09	Not tested / Open	_____
A17-10	Not tested / Open	_____
A17-11	Not tested / Open	_____
A17-12	Not tested / Open	_____
A17-13	Not tested / Open	_____
A18-01	Not tested / Open	_____
A18-02	Not tested / Open	_____
A18-03	Not tested / Open	_____
A18-04	Not tested / Open	_____

Closed actions: None pre-populated. A17-01 to A17-13 remain open, alongside A18-01 to A18-04. Reported Arduino/relay registration is partial evidence only. Open deviations: _____. Reviewer/name/date: _____. Accepted article/configuration/limits: _____.

CAUTION
Current disposition: combined R18 prototype review with manufacturing, physical and electrical acceptance open. Each signed result applies only to the identified article and recorded criteria. Neither reported board fit nor digital geometry checks establish leak, power, ingress or unattended-operation acceptance.